

Support theorems and duality for relative Hitchin systems

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Relative Higgs bundles for X_S and X_U

- Σ smooth, projective curve of genus $g \geq 2$.
- D a line bundle on Σ of degree $\geq 2g$
- $\vartheta: \Sigma' \rightarrow \Sigma$ unramified, étale double cover
- Ngô and Laumon (2004) studied “unitary Higgs bundles”:

$$\mathrm{Higgs}_{U_{2n}} := \left\{ (\mathcal{E}, \phi, h) \left| \begin{array}{l} \mathcal{E} \text{ vector bundle on } \Sigma' \\ \phi: \mathcal{E} \rightarrow \mathcal{E} \otimes \vartheta^* D \\ h: \mathcal{E} \rightarrow \sigma^* \mathcal{E}^\vee \text{ Hermitian, i.e. } h = \sigma^* h^\vee \end{array} \right. \right\}$$

- We have involutions ι_S and ι_U on $\mathrm{Higgs}_{U_{2n}}$ by

$$\iota_S(\mathcal{E}, \phi, h) = (\mathcal{E}, -\phi, -h), \quad \iota_U(\mathcal{E}, \phi, h) = (\mathcal{E}, -\phi, h)$$

- Put $\mathrm{Higgs}_\star = (\mathrm{Higgs}_{U_{2n}})^{\iota_\star}$ for $\star \in \{S, U\}$.

Relative Hitchin Fibration for X_S and X_U

- Define $f_{U_{2n}}: \text{Higgs}_{U_{2n}} \rightarrow \mathcal{A}_{U_{2n}}$ by $f_{U_{2n}}(E, \phi, h) = \text{char.poly}(\phi)$.
- $f_{U_{2n}}$ induces maps

$$\begin{array}{ccc} \text{Higgs}_S & & \text{Higgs}_U \\ & \searrow f^S & \swarrow f^U \\ & \mathcal{A} & \end{array}$$

- For $a = (a_2, a_4, \dots, a_{2n}) \in \mathcal{A}(k)$, can form

$$\tilde{\Sigma}'_a = \{\lambda^{2n} + a_2 \lambda^{2n-2} + \dots + a_{2n} = 0\} \subset \text{Tot}(\vartheta^* D)$$

with involutions σ (Frobenius on coefficients + $\lambda \mapsto -\lambda$) and ι ($\lambda \mapsto -\lambda$).

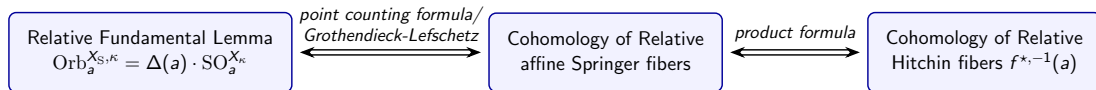
- If $\tilde{\Sigma}_a$ is reduced,

$$f^{*, -1}(a) \simeq \{(\mathcal{L}, h, \xi) \mid \mathcal{L} \in \overline{\text{Pic}}(\tilde{\Sigma}'_a), h: \mathcal{L} \rightarrow \sigma^* \mathcal{L}^\vee, \xi: \mathcal{L} \simeq \iota^* \mathcal{L}\}$$

When $\tilde{\Sigma}_a$ is smooth, this is a union of $\mathcal{P}_a$ torsors, for the abelian variety

$$\mathcal{P}_a \simeq \{(\mathcal{G}, \xi) \mid \mathcal{G} \in \text{Pic}(\tilde{\Sigma}'/\iota), \xi: \mathcal{G} \rightarrow \sigma^* \mathcal{G}^{\otimes -1}\} = \text{Prym}(\tilde{\Sigma}'_a/\iota, \tilde{\Sigma}'_a/\langle \iota, \sigma \rangle)$$

Geometric Stabilization



- For an element $\kappa \in H^1(k, \mathcal{P}_a)^D$ [tuple of ± 1 's], we get:
 1. an endoscopic symmetric space $X_\kappa = X_S^r \times X_U^s$ where $r + s = n$ denote ranks;
 2. a canonical transfer map $\nu_\kappa: \mathcal{A}_\kappa \rightarrow \mathcal{A}$;
 3. a decomposition of cohomology $Rf_*^* \overline{\mathbb{Q}}_\ell \simeq \bigoplus_\kappa (Rf_*^* \overline{\mathbb{Q}}_\ell)_\kappa$.

Geometric Stabilization for Unitary Friedberg-Jacquet (H.-Leslie)

There is an isomorphism

$$\nu_\kappa^* (Rf_*^S \overline{\mathbb{Q}}_\ell)_\kappa \simeq (Rf_*^\kappa \overline{\mathbb{Q}}_\ell)_{\text{st}}.$$

up to Tate twist/cohomological shift. [Over an anisotropic locus; base changed to a cameral cover.]

Proof Sketch of Geometric Stabilization

Ngô's Support Theorem

Suppose that we have a δ -regular weak abelian fibration

$$M \xrightarrow{f} A \xleftarrow{g} P$$

with M smooth and f proper and flat. If Z is a support of simple constituent of $Rf_*\overline{\mathbb{Q}}_\ell$, then:

1. Z is “ δ critical”;
 2. there must exist an open set $U \subset A$ intersecting Z such that $R^{\text{top}}f_*\overline{\mathbb{Q}}_\ell|_{U \cap Z}$ is a local system and is a direct factor of $R^{\text{top}}f_*\overline{\mathbb{Q}}_\ell|_U$.
- Paraphrasing: It is enough to compute top *ordinary* cohomology over dense, open sets of “ δ critical” loci.

Regular Locus in $\text{Higgs}_\star$

- Let $\mathfrak{u}_{2n} = \text{Lie}(U_{2n})$.
- We have symmetric subgroups $H_\star = \begin{cases} \text{GL}_{n,E} & \text{if } \star = S \\ U_n \times U_n & \text{if } \star = U \end{cases}$ cut out by involutions $\theta_\star$.
- The orthogonal complement of $\mathfrak{h}_\star = \text{Lie } H_\star$ is

$$\mathfrak{h}_\star^\perp := \mathfrak{u}_{2n}^{\theta_\star = -1} = \begin{cases} \text{Herm}_n \times \text{Herm}_n & \text{if } \star = S \\ \mathfrak{gl}_{n,E} & \text{if } \star = U \end{cases}$$

Definition

Let $\mathfrak{h}_\star^{\perp, \text{reg}}$ be the set of $x \in \mathfrak{h}_\star^\perp$ such that $\dim H_\star \cdot x$ is maximal.

$\text{Higgs}_\star$ is the moduli of tuples $(\mathcal{E}, \phi, h, \xi) \in \text{Higgs}_\star$ such that ϕ is valued in $\mathfrak{h}_\star^{\perp, \text{reg}}$.

- In fact, $\text{Higgs}_\star \simeq \text{Sect}_\Sigma([(\mathfrak{h}_\star^\perp)_D / H_\star])$, and $\text{Higgs}_\star^{\text{reg}} \subset \text{Higgs}_X$ denotes maps that factor through $[(\mathfrak{h}_\star^{\perp, \text{reg}})_D / H_\star]$.

Geometry over the regular locus

Theorem (H.-Leslie)

There is a nonempty open set $U \subset \mathcal{A}^{\text{ani}}$ such that:

1. For every $a \in U(k)$, $\text{Higgs}_{\star}^{\text{reg}} \cap f^{\star, -1}(a)$ is dense in $f^{\star, -1}(a)$;
2. For every irreducible, δ -critical set $Z \subset \mathcal{A}^{\text{ani}}$, $U \cap Z \neq \emptyset$.

Theorem (H.-Morrissey)

$\text{Higgs}_{\star}^{\text{reg}}$ is a torsor for a smooth, δ regular (generically abelian) group scheme **over an étale cover $\mathcal{A}^{\text{reg}}$** of $\mathcal{A}$.

- The cover $\mathcal{A}_X^{\text{reg}} \rightarrow \mathcal{A}_X$ is a space of twisted maps into the cover

$$\mathfrak{t} := \{\text{Regular } H \text{ orbits in } \mathfrak{h}^\perp\} \rightarrow \{\text{Closed } H \text{ orbits in } \mathfrak{h}^\perp\} =: \mathfrak{c} \quad (1)$$

Examples

Consider the $n = 1$ case over $k = \bar{k}$. $X_\star \otimes_k \bar{k} \simeq \text{GL}_2/\mathbb{G}_m \times \mathbb{G}_m$. Then $H = \mathbb{G}_m \times \mathbb{G}_m$ acts on $\mathfrak{h}^\perp \simeq \mathbb{A}^2$ by the hyperbolic action $\alpha \cdot (x, y) = (\alpha x, \alpha^{-1} y)$. We have $\mathfrak{h}^{\perp, \text{reg}} = \mathbb{A}^2 \setminus \{0\}$. In this case, H acts freely on $\mathfrak{h}^{\perp, \text{reg}}$, and (1) is

$$\text{---} \overset{\cdot}{\text{---}} \rightarrow \text{---}$$

Frobenius acts on the cover by swapping the doubled points in the unitary case and fixing both in the Siegel case.

Fibers of the map $f^{\star, \text{reg}}: \text{Higgs}_\star^{\text{reg}} \rightarrow \mathcal{A}$ are lifts of the curve over this map.

Generalizations of $\text{Higgs}_\star$

- Let G be a reductive group and X a *spherical variety*.
- We assume that X has no “type N roots.”
- For simplicity, we also assume that $X = G/H$ is *homogeneous* and *affine*.
- Examples:
 - Symmetric varieties $X = G/G^\theta$
 - “Tate” $\mathbb{A}^1$ with $\mathbb{G}_m$ action
 - “Rankin-Selberg” $\text{GL}_n \times \text{GL}_{n+1}/\text{GL}_n$
 - “Gross-Prasad” $\text{SO}_n \times \text{SO}_{n+1}/\text{SO}_n$

Definition

Fix Σ a smooth, projective curve and D a line bundle on Σ of degree $\geq 2g$. Then

$$\text{Higgs}_X := \text{Sect}([(T^*X)_D/G]) = \text{Sect}([\mathfrak{h}^\perp)_D/H])$$

X Higgs bundles

- There is a canonical morphism

$$\mathrm{Higgs}_X \rightarrow \mathrm{Higgs}_G$$

induced by the moment map $T^*X \rightarrow \mathfrak{g}^*$.

- When $D = \omega_\Sigma$, Higgs_X is the *conormal bundle* to Bun_H in Bun_G , and the image of Higgs_X is a *Lagrangian* in $\mathrm{Higgs}_G = T^*\mathrm{Bun}_G$.
- Still have a fibration $f_X: \mathrm{Higgs}_X \rightarrow \mathcal{A}_X$ for an affine space $\mathcal{A}_X$.

Failures of regularity

- f_X is not flat in general; Ngô's Support Theorem is not sufficient!
- Generic fibers of f_X need not be torsors under an abelian variety; e.g. can look like moduli of rank 2 vector bundles on a family of spectral curves.

Definition/Proposition

X is *tempered* if the generic centralizers of G acting on T^*X (or equivalently, H acting on $\mathfrak{h}^\perp$) are abelian.

For tempered X , generic fibers of f_X are unions of torsors for an abelian variety.

- Luckily, regularity is enough to study *smooth* fibers for tempered spherical varieties.
- Not enough for fundamental lemmas, but enough for *duality*.

Classical limit of geometric Langlands

- Let $\widehat{G}$ be the Langlands dual group of G .
- $\widehat{M}$ the [BZSV] dual Hamiltonian of $M = T^*X$, determined by a group $\widehat{G}_X \subset \widehat{G}$ and a symplectic representation S_X of $\widehat{G}_X$.

$\mathcal{P}_X^{\text{dR}}$: [BZSV] Period sheaf

$\mathcal{L}_{\widehat{M}}^{\text{dR}}$: [BZSV] L sheaf

$$\begin{array}{ccc}
 D\text{-mod}(\text{Bun}_G) & \xleftarrow[\text{Geometric Langlands}]{\sim} & \text{IndCoh}(\text{Loc}_{\widehat{G}}) \\
 \left. \begin{array}{c} \text{associated graded} \\ \text{of a filtration} \end{array} \right\} \downarrow & & \downarrow \left. \begin{array}{c} \text{"nonabelian Hodge"} \end{array} \right\} \\
 \text{Coh}(\text{Higgs}_G) & \xleftarrow[\text{Fourier-Mukai, FM}]{\sim} & \text{Coh}(\text{Higgs}_{\widehat{G}})
 \end{array}$$

$$\mathcal{P}_X = \mu_* \mathcal{O}_{\text{Higgs}_X}$$

$$\mathcal{L}_{\widehat{M}} = \text{DH}(S_X, \widehat{G}_X): \text{Dirac-Higgs bundle}$$

- Assume X is tempered and $\mathrm{Higgs}_X \rightarrow \mathrm{Higgs}_G$ intersects the smooth locus of Higgs_G .

Theorem (H.-Luo-Morrissey)

The identity $\mathrm{FM}(\mathcal{P}_X) = \mathcal{L}_{\hat{M}}$ holds “up to translations” assuming the dual data $(\hat{G}_X, S_X)$ satisfies *divisor matching*.

- “Up to translations” related to Hecke-Wilson compatibility when $\hat{M}$ is polarized (future work)

Divisor Matching

Conjecture (H-Luo-Morrissey); Theorem announced by [BZSV] for X strongly tempered

There is a natural identification

$$\mathfrak{h}^\perp // H \simeq \widehat{\mathfrak{g}}_X // \widehat{G}_X$$

and under this identification, the divisor $\mathfrak{D}_{S_X} = (\sqrt{\det(S_X)})$ is equal to

$$\mathfrak{D}_{\text{br}} = \overline{\{x \in \mathfrak{h}^\perp // H \mid \text{there are exactly 2 regular } H \text{ orbits above } x\}}$$

when $\mathfrak{D}_{\text{br}}$ is a divisor, and $S_X = \text{triv}$ if $\mathfrak{D}_{\text{br}}$ is not a divisor.

- We prove the above for a number of examples of spherical varieties.
- This result can be used to *define* S_X conditioned on a conjecture in [BZSV] that S_X always have miniscule weights.